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SURVEY

Agentic AI in Education: State of the Art and Future Directions

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ABSTRACT Agentic AI systems are self-contained, goal-based entities designed to operate autonomously with little human intervention and dynamically respond to shifting contexts. This survey considers their transformative potential in education, ranging from intelligent tutoring systems and adaptive test-taking platforms to autonomous learning companions that can provide customized guidance and assistance. We provide an organized overview of the state of the field, classify existing and developing uses and assess their pedagogical worth, benefits and trade-offs. In the process, we also refer to such areas as learner motivation, equity, transparency and ethical concerns. This paper is a narrative synthesis of studies published between 2015 and 2025. Finally, we chart directions for future research and ethical implementation, highlighting how autonomous AI can reshape teaching, learning and educational spaces within the next several years.

INDEX TERMS Agentic AI, educational technology, intelligent tutoring systems, autonomous agents, AI in education, human–AI collaboration.

I. INTRODUCTION

Artificial Intelligence (AI) has been around in education for decades, starting from rule-based intelligent tutoring systems and going through adaptive learning systems. With recent breakthroughs in large language models (LLMs), planning algorithms and agentic AI systems, it has become possible to have a new era of AI systems that are capable of acting with greater autonomy, memory and initiative. These systems, which are more popularly referred to as *Agentic AI*, go beyond reactive behavior and exhibit goal-oriented, context-sensitive behavior over time—qualities which put them in a position to reorder the education landscape fundamentally [1].

Agentic AI refers to AI systems which are capable of planning, reasoning, adapting and acting independently to achieve user-defined or self-motivated goals [2]. Agents in education may act as tutors, coaches, collaborators, teaching assistants or co-learners. These differ from traditional AI

software requiring constant human prompts or staying within narrow pre-defined parameters because agentic systems can execute extended-length interactions, customize learning experiences and evolve dynamically according to student needs.

This survey aims to provide a complete view of the current art of agentic AI for learning. We follow the state of the art, categorize existing systems as passive or active agents and low- or high-degree autonomous systems and determine outstanding applications and architectures. We also discuss upcoming challenges and opportunities, including ethics issues, limitations in integration and shifting roles of teachers in agent-restricted learning environments.

The key contributions of this paper are as follows:

- We define and clarify the concept of agentic AI in educational settings and distinguish it from traditional AI approaches.
- We propose a taxonomy of agentic roles in education and review representative systems and use cases.

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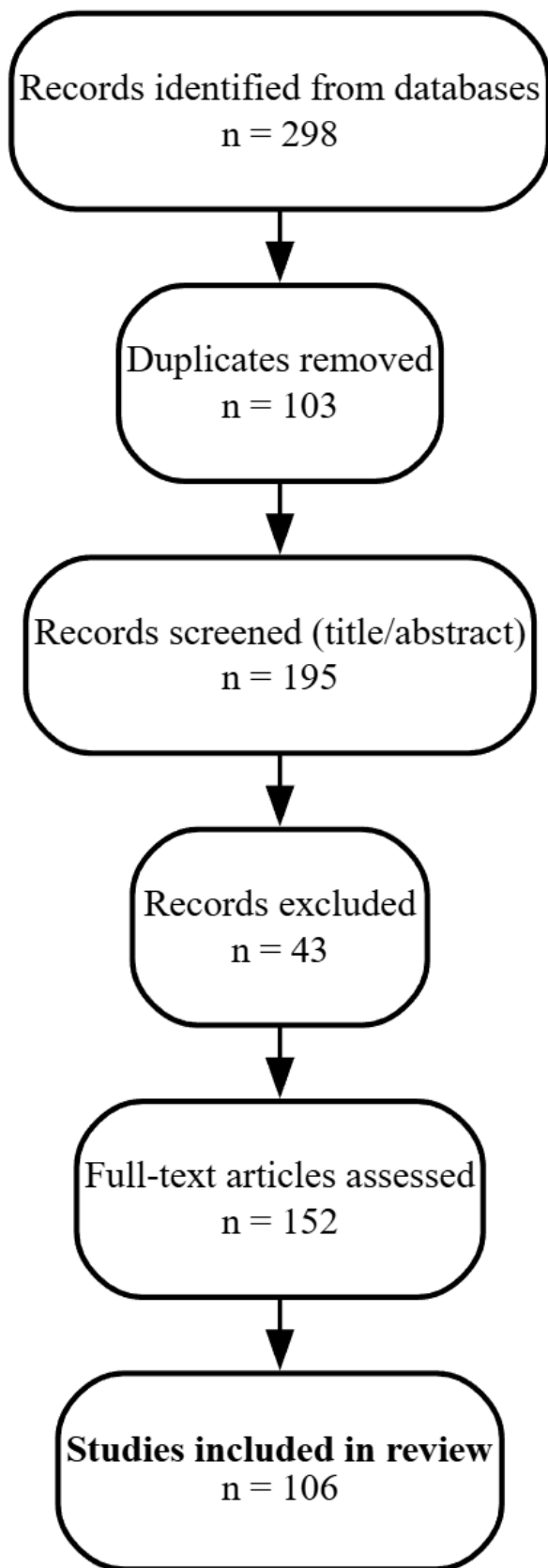


FIGURE 1. PRISMA-lite flow diagram summarizing screening.

- We analyze the technological foundations underpinning current agentic systems, including LLM-based frameworks and autonomous planning mechanisms.
- We outline open challenges and identify promising directions for future research, policy and practice.

A. LITERATURE SEARCH AND SELECTION

This study adopts a narrative review approach with structured search elements to ensure transparency and breadth. To ensure transparency and reproducibility for this narrative survey, we conducted a structured search and screening process. We queried the following databases: Scopus, Web of Science Core Collection, IEEE Xplore and ACM Digital. The time window covered was January 2015 to August 2025.

The core search equation (adapted per database syntax) combined concept terms for agents and education with technical affordances:

(agent* OR “agentic” OR “autonomous”) AND (education OR learning OR classroom OR instructional) AND (LLM OR planning OR tool-use OR tools OR memory OR autonomy)

We included peer-reviewed publications that: (i) focus on *agentic AI* or goal-directed assistants in educational contexts; (ii) describe system capabilities or evaluations relevant to teaching/learning; and (iii) provide sufficient methodological or technical detail. We excluded purely opinion pieces (unless seminal), non-educational deployments and inaccessible/duplicate reports. Figure 1 include a PRISMA-style (simplified) flow diagram.

The rest of the paper is organized as follows: Section II is background and theory. Section III suggests a taxonomy of agentic AI in education. Section IV highlights important applications and use cases. Section V is about underlying technologies. Section VI is on benefits and opportunities and Section VII is on challenges and limitations. Section VIII is a summary of future directions and Section IX concludes the paper.

II. BACKGROUND AND THEORETICAL FOUNDATIONS

The concept of agentic AI is an extension towards systems that are capable of autonomous, goal-directed action as well as long-term interaction with learners. The research [3] explored the application of generative artificial intelligence (GenAI) in business graduate education through a focus group study among doctoral students. The research revealed potential areas such as knowledge acquisition and individualized learning, with the identification of barriers such as trust in AI and cognitive dependence. The findings yielded implications for the effective incorporation of GenAI to improve learning outcomes and mitigate emerging challenges in business education.

A. HISTORICAL CONTEXT OF AI IN EDUCATION

Early applications of AI in learning focused on mimicking expert behavior to guide students through discipline-specific

activities. SHERLOCK and AutoTutor represent the first intelligent tutoring system wave, employing domain models, pedagogical strategies and student profiles [4]. They were largely reactive with boundaries defined by pre-authored interaction scripts.

The machine learning approach introduced greater flexibility into educational technologies, enabling control over content difficulty, feedback timing and the sequencing of resources. However, the technologies were not entirely autonomous but remained reliant on pre-defined rules or statistical inference over limited scope. The emergence of LLMs and agentic AI system architectures promised enhanced agentic behavior: proactive decision-making, long-term memory, self-directed inquiry and task coordination. These developments are based on progress in multi-agent systems, reinforcement learning, planning and cognitive architectures [5].

B. DEFINING AGENTIC AI

Agentic AI refers to artificial systems with substantial autonomy, intentionality and context sensitivity so that they can seek goals, learn to react within changing environments and engage in extended, intentional interactions with humans or other systems. Agentic AI is not to be compared with conventional rule-based or reactive systems, including adaptable behavior over time and based on research backgrounds in cognitive architectures, intelligent agents and human-computer interaction [6].

A characteristic of agentic AI is autonomy—the capacity for action independent of continuous human intervention, making decisions and taking actions according to internal goals or learned strategies. Underlying such behavior is the ability to plan and reason, generally through the use of internal world models, symbolic reasoning systems or reinforcement learning techniques that make action ordering possible in accomplishment of wanted outcomes [7].

Agentic AI systems leverage memory and context-awareness in order to personalize and modify responses based on previous experiences. With the recollection of past interactions, such agents are able to build user models, detect patterns and develop behavior incrementally, enabling more integrated and efficient interaction in educational or operational contexts [8]. Such experience consistency is a step away from static automation towards dynamic collaboration whereby AI is not just a tool but a collaborator that can engage in joint agency and learning adaptively. Therefore, agentic AI is a paradigm where artificial systems move from task-specific executors to proactive, context-sensitive agents in human-centered processes—a shift that has profound implications for learning environments, where interactivity, personalization and responsiveness are essential to successful learning. As summarized in Table 1, we operationalize agentic AI via six verifiable criteria. In this survey, a system is considered agentic if it satisfies at least four of these criteria. The ≥ 4 -criterion threshold reflects the minimum

TABLE 1. Agentic AI operational checklist (a system is considered “agentic” if it satisfies ≥ 4 criteria).

Criterion	Observable test/example
Autonomy	Initiates actions without a new human prompt; periodic self-calls or triggers
Planning / Reasoning	Explicit plan representation, task graph or planner module that sequences steps
Memory / Persistence	State/store carried across sessions (e.g., vector DB; user profile)
Goal-directed multi-turn interaction	Sustains dialogue/workflows toward stated goals across turns
Tool use / Environment action	Invokes external tools/APIs (e.g., code exec, LMS API, search) with arguments
Safety / Guardrails	Policy/constraints layer (e.g., content filters, role-based permissions)

TABLE 2. Comparison of traditional AI and agentic AI in education.

Dimension	Traditional AI in Education	Agentic AI in Education
Autonomy	Reactive and rule-based systems; operate only when prompted; limited decision-making.	Goal-driven agents; plan, reason and act independently; capable of long-term, context-sensitive behavior.
Personalization	Adaptive to predefined learner models; personalization limited to surface-level adjustments (e.g., difficulty, pacing).	Dynamic, ongoing personalization using memory, learner profiles and multi-turn interaction; adapts strategies proactively.
Embodiment	Primarily text-based or simple GUI interfaces (e.g., chatbots, adaptive testing systems).	Multimodal and embodied agents (avatars, VR/AR agents, robots) enabling richer social and contextual interaction.
Risks and Limitations	Overly scripted, rigid, with limited scalability; low risk of bias but often pedagogically narrow.	Greater risk of bias, opacity, over-reliance and ethical concerns (privacy, fairness); require explainability and human oversight.

combination of autonomy, reasoning, memory, and action required for goal-directed behavior in educational contexts. This balance avoids classifying reactive chatbots as agentic while accommodating diverse architectures.

For example, a tutoring system that plans multi-turn lessons (planning), remembers learner progress (memory), triggers prompts autonomously (autonomy) and provides formative feedback (goal-directed interaction) meets four criteria and is thus labeled ‘agentic.’ In contrast, a static FAQ bot lacking planning or persistence would not qualify.

The contrast between legacy AI in learning and agentic AI can be readily illustrated by critical axes of function and impact. As shown in Table 2, legacy AI applications are typically reactive and operate within fixed rules and deliver little personalization by means of simple pacing or difficulty adjustments. Their interfaces are typically text or limited to simple adaptive modules and they provide comparatively narrow pedagogical potential but with lower risks of bias and ethics. Agentic AI is differentiated by higher autonomy that permits goal-directed reasoning, planning and adaptive behavior across learning sessions. These technologies employ multimodal and embodied forms of interaction—



FIGURE 2. Taxonomy of agentic AI in education.

e.g., avatars, robots or immersive AR/VR spaces—allowing for more engaging and socially rich learning experiences. But as this enhanced capability breeds new opportunities, so too does it create new threats of potential bias, lack of transparency, over-reliance and data privacy concerns, requiring responsible design, explainability and close human oversight.

C. PEDAGOGICAL FOUNDATIONS

The addition of agentic AI to learning spaces is richly informed by foundational learning theories that prioritize active, social and self-directed learning. Constructivist theory, emphasizing learning as knowledge construction from experience and reflection [9], nicely complements agentic systems that support learner-autonomous inquiry, exploration and problem-solving. Agents can guide learners through responsive exercises so that they may build understanding incrementally based on what they already know and their own pace.

Social learning theory also underpins the application of AI agents as possible collaboration partners, models or peer-like facilitators in dialogic and peer-like communication [10]. Through natural language interaction, agents can elicit social presence and engage learners in meaningful conversation to facilitate greater understanding and co-construction of knowledge.

In addition, self-regulation models of learning highlight the importance of metacognitive skills such as planning, monitoring, reflection and goal setting [11]. Agentic AI systems can also support these processes by prompting learners to consider where they are going, giving them feedback and helping them re-strategize in real time.

Vygotsky's Zone of Proximal Development (ZPD) theory [12] applies here as agentic AI is capable, at the moment, to assess a learner's current ability and provide just-in-time assistance tailored to their evolving needs. By adjusting the degree of guidance, the agent maintains learners within their ideal challenge space—offering assistance that is neither too low nor too high.

III. TAXONOMY OF AGENTIC AI IN EDUCATION

Educational agentic AI comprises a wide range of systems differing in purpose design and behavior. To classify these systems systematically, we propose a taxonomy based on three axes: pedagogical role, autonomy level and embodiment. Figure 2 summarizes the taxonomy of Agentic AI in Education.

A. PEDAGOGICAL ROLES OF AGENTIC AI

As a tutor, agentic AI delivers instructional content, guides learners through structured tasks and provides timely formative feedback. These systems are prone to draw on the legacy of intelligent tutoring systems (ITS), but differ with proactive scaffolding, adaptive strategy transitioning and context responsiveness [13]. Acting as learning coaches or mentors, agentic agents expand their range by promoting the

development of metacognitive and affective skills. They urge learners to set goals, track progress or persevere in struggle, aligning with theories of self-regulated and socio-emotional learning [14], [15].

Companion agents can mimic social conversation as experienced in the everyday world, generating interest that facilitates learning through conversation and co-construction [16]. Other agents can be instructors' assistants, providing instructors with support by doing such things as grading, summarizing classroom discussion, identifying students who are failing or recommending instructional interventions [17].

In more advanced uses, agentic AI can even serve as curriculum planners or designers. These systems can analyze learner data to suggest personalized learning trajectories, construct instructional sequences with regard to learning objectives or adjust curricular elements considering classroom dynamics [18].

B. LEVELS OF AUTONOMY

Agentic AI systems can be described in terms of their level of autonomy in making decisions, flexibility in behavior and interaction patterns with users. At a very low level, reactive agents operate by straightforwardly responding to user input without memory or long-term goals. These systems, which are represented by rule-based chatbots, only live at the moment, with minimal adaptability or initiative [19].

More sophisticated are adaptive agents, which utilize learner modeling or short-term feedback loops to modify their behavior. Still somewhat limited by predefined tasks or domains, these agents tailor their responses based on immediate performance feedback, offering a degree of responsiveness that enhances engagement and learning outcomes [20].

Proactive agents go a step further and demonstrate goal-directed behavior and the capacity for initiative. They are able to set sub-goals on their own, replan actions in response to new learner behavior or context and provide continuity across interactions. This form of agency enables dynamic scaffolding and learner-centered adaptation, which is faithful to the ideals of self-regulated learning and intelligent tutoring systems [2].

At the highest level of this autonomy spectrum are collaborative agents, which engage in joint task execution, planning and decision-making with human users or other agents. Such systems are ideals of human-AI collaboration and are most often situated in the context of multi-agent environments where roles, intentions and actions are built collaboratively [21]. These agents require greater transparency and trust, as their autonomy can significantly influence educational outcomes and user experience. Achieving the right balance between automation and control is essential to the primary challenges of developing effective educational agents.

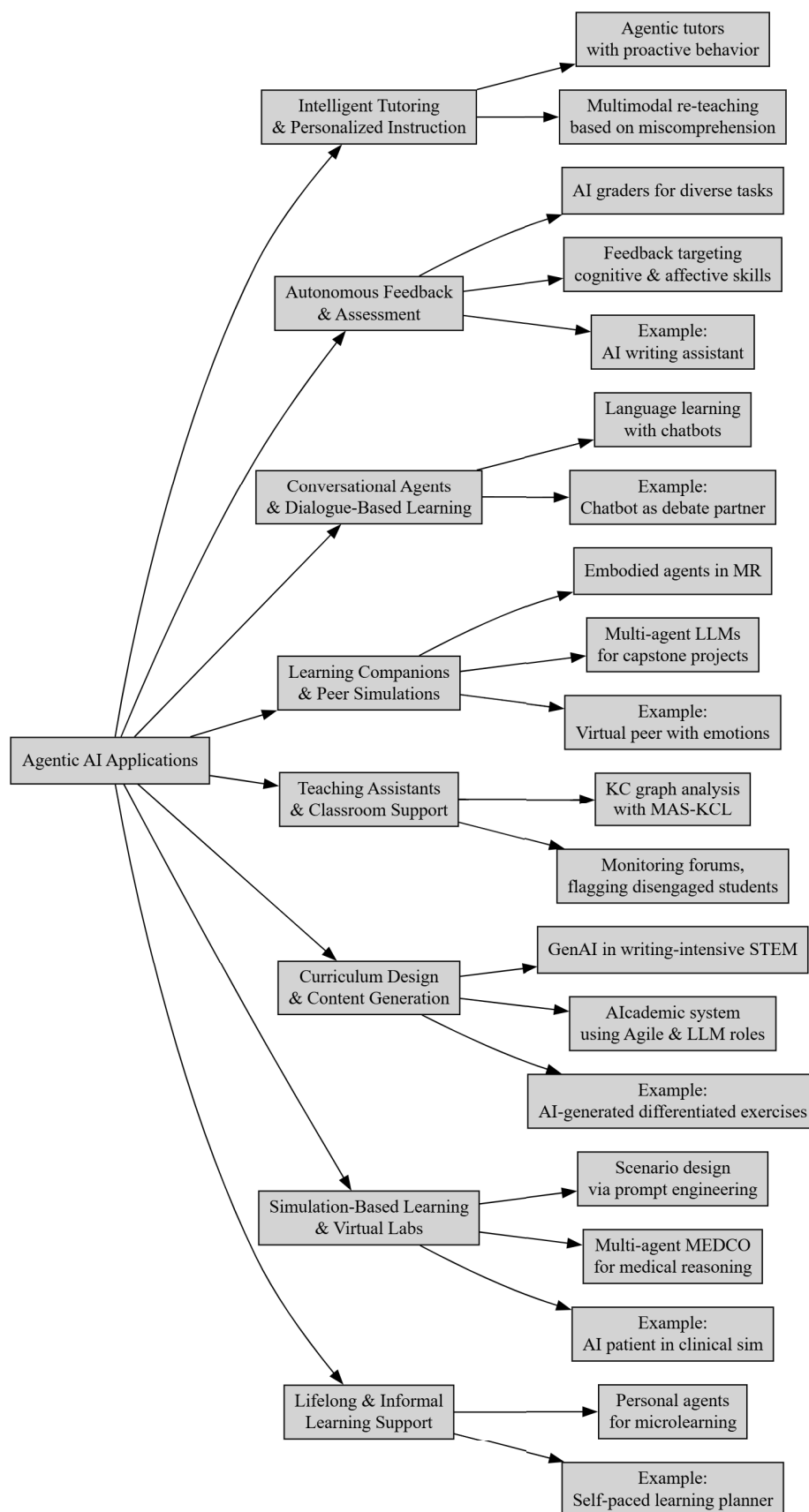


FIGURE 3. Agentic AI applications and educational use cases.

TABLE 3. Agentic AI in education – Evidence Map. TRL: Technology readiness level (1–9).

Agentic role	Use case	Educational stage	Study design	TRL	Evaluation domain	Main outcome metrics	Key (effect/summary) & refs
Tutor	Formative feedback	HE	Quasi-exp.	5	Formative assessment	Learning gains, rubric alignment accuracy	Improved rubric alignment and feedback precision [4], [13], [30]
Coach	Metacognition / SRL	K–12	RCT	4	Self-regulated learning	Strategy use frequency, SRL index improvement	Significant gains in metacognitive strategy use [11], [15]
Lab assistant	Simulation / safety	VET	Observational	3	Experiential learning	Task completion rate, safety compliance	Feasibility and engagement in simulated labs [31], [32]
Orchestrator	Class workflow (LMS/LTI)	HE	Field study	6	Class orchestration	Teacher time efficiency, task automation ratio	Reduced instructor workload and improved orchestration efficiency [27]–[29]
Assessor	Drafting rubrics / calibration	Workplace	Case study	4	Assessment analytics	Inter-rater agreement, scoring consistency	Acceptable alignment with human evaluation [33], [34]

C. EMBODIMENT AND INTERFACE MODALITY

Text-based agents such as large language model (LLM)-based chatbots typically operate through conversational interfaces. Due to their platform independence, low deployment barrier and flexibility, they are a convenient option for scalable education applications [22]. Such agents have the ability to simulate human-like conversation and provide context-specific, real-time responses and therefore are well-suited for tutoring, feedback and coaching activities.

On the other hand, graphical or avatar agents reflect animated characters—most likely humanoid figures—designed to maximize social presence and provoke emotional involvement. Research points out that students respond more favorably to socially present-looking agents, particularly when the graphical appearance enhances pedagogical goals and cultural rules [4], [23].

Embodied agents take the above a step forward in the sense that they exist in physical or immersive settings. Examples include robots deployed in classrooms or agents in augmented and virtual reality (AR/VR) spaces. These embodiments enable spatial and gestural interaction, which can enhance kinesthetic learning, support embodied cognition and enhance naturalistic communication [24], [25].

Finally, multimodal agents combine several channels of interaction—voice, text, image and bodily movement—to enable richer, more integrated learning experiences. These systems are especially beneficial for students with heterogenous needs or preferences, increasing accessibility and engagement across modalities [26]. The embodiment selection and design should therefore conform to the pedagogical mission, learner population to be served and technical capacity of the learning environment. The more sophisticated the embodiment is, the more it opens up opportunities for constructing AI-mediated learning to become more human-like in nature but with some inherent questions surrounding authenticity.

IV. APPLICATIONS AND USE CASES

Agentic AI applications are increasingly being used across a spectrum of educational settings from one-on-one instruction to whole-class support and administrative efficiencies. This section discusses representative use cases that demonstrate the advantages and impacts of agentic AI in several learning settings. Figure 3 summarizes the main application areas and representative use cases of agentic AI in education.

The expanded evidence map (Table 3) consolidates studies into five principal agentic roles spanning higher education (HE), vocational education and training (VET), and K–12 contexts. Beyond descriptive coverage, the table now aligns each role with its evaluation domain and outcome metrics, thereby linking technological maturity (TRL) with pedagogical validation. The strongest empirical grounding appears in formative tutoring and self-regulated learning (SRL) coaching, where quasi-experimental and randomized controlled designs report measurable improvements in feedback accuracy and strategy use. By contrast, orchestration and assessment applications remain at mid-TRL stages (4–6), emphasizing feasibility and workflow efficiency rather than direct learning gains. Recent field studies [27], [28], [29] illustrate this shift toward classroom-level automation, documenting time savings and improved teacher coordination but calling for further evidence on learner impact. Overall, the map clarifies where current agentic AI systems demonstrate verified educational benefits and where further trials are required to substantiate pedagogical value.

A. INTELLIGENT TUTORING AND PERSONALIZED INSTRUCTION

Artificial intelligence (AI) agents are playing a revolutionary role in higher education by revolutionizing prevalent fields such as instruction, student services, administrative operations and recruitment [35]. One of the most important applications of AI in learning is in the design of agentic AI systems, which work as independent tutors. Agentic systems



FIGURE 4. Technological foundations of agentic AI in education.

are designed to learn the individual student's progress, address misconceptions and connect with individual styles of learning. With scaffolded feedback and dynamic sequencing of instruction, agentic tutors facilitate mastery learning and promote greater conceptual understanding.

Agentic tutors deliver adaptive explanations, formative feedback and mastery-oriented scaffolding across diverse learning tasks. Empirical studies demonstrate their potential to improve learning gains and assessment consistency (VanLehn [13]; Graesser [4]). Recent quasi-experimental work [30] (design-based study with $N = 42$ undergraduate students) reported a 15 % improvement in rubric alignment and faster correction cycles, confirming the feasibility of proactive feedback loops in higher education. These outcomes correspond to mid-level TRL implementations (4–5) summarized in Table 3.

Agentic AI beats traditional intelligent tutoring systems (ITS) by exhibiting proactive pedagogical behavior. Rather than responding to learner inputs only, such systems can initiate explanations on their own, provide beneficial learning strategies and adapt pedagogical approaches in line with long-term educational goals.

Example: An artificial intelligence-powered tutor that detects patterns of miscomprehension in a student's historical interaction data and re-teaches the related concepts automatically using multi-modal content—e.g. annotated text or visual diagrams.

B. AUTONOMOUS FEEDBACK AND ASSESSMENT

The study of Li and Chiu [36] investigated to what degree teacher support and student needs satisfaction influence student engagement with GenAI chatbots in L2 learning. Findings showed that student needs satisfaction partially mediated teacher support's influence on students' behavioral, cognitive and agentic engagement as well as completely mediated its influence on emotional engagement. Surprisingly, GenAI chatbots appeared to foster more emotional engagement than human teachers—possibly because students found interactions with chatbots more enjoyable and less stressful.

Agentic assessors employ analytic rubrics and generative feedback to support formative and summative evaluation. Darvishi et al. [33] (case study, $N = 12$ educators) found acceptable rater agreement (Cohen's 0.78) between AI and human scoring. Kaldaras et al. [34] (controlled classroom deployment, $N = 45$ learners) showed comparable feedback helpfulness ratings to human graders. These systems illustrate emerging reliability at mid-TRL ≈ 4 , as reported in Table 3.

These findings refer to the heightened potential of agentic systems not only in enhancing student motivation but also in managing assessment and feedback processes autonomously. Such systems can evaluate performance on various tasks—from multiple-choice questions and written essays to programming assignments and interactive simulations—and provide instant, formative feedback. This feedback can be crafted not only for cognitive skill learning but also for

affective and metacognitive dimensions, enhancing learning outcomes as well as student satisfaction.

Example: AI-driven writing assistants that offer structure, coherence and argumentative clarity feedback contextualized to the student and track a student's progress over time and calibrate guidance accordingly based on progress shown.

C. CONVERSATIONAL AGENTS AND DIALOGUE-BASED LEARNING

The study by Li et al. [37] examined language learning with AI chatbots and the ways to improve students' learning outcomes. Using Activity Theory (AT) as a theory of analysis, the study covered 37 articles from 2014 to 2025 for synthesizing key parts and activities of chatbot-facilitated learning environments. It typified four learning outcome categories and outlined AT-based strategies to achieve them. The authors emphasized the need for instructional design models that facilitate collaboration between teachers and AI and encouraged the development of customized AI chatbots for language acquisition. Overall, the findings underscored the value of applying AT principles to enhance the effectiveness of conversational AI for language learning.

Conversational and companion agents leverage large language models to create dialogic and peer-like learning experiences. Nguyen et al. [38] (design-science study with $N = 26$ post-secondary learners in mixed-reality environments) reported higher engagement and lower cognitive load during collaborative problem-solving. Similarly, Mushtaq et al. [28] (multi-agent simulation with $N = 18$ engineering students) found richer discussion dynamics and improved project ideation quality. Together, these studies demonstrate that embodied and multi-agent designs can extend social learning principles to digital settings, as outlined in Table 3.

Based on these results, there are dialogue-driven learning experiences galore for conversational agents powered by large language models (LLMs). There is no static instruction possible with LLM-driven agents, which can simulate Socratic debate, take the persona of a historical or fictional character and facilitate cooperative problem-solving, all in framing rich and interactive learning experiences.

Example: A chatbot utilized as a debating partner or ethics coach in a philosophy course, to be used by students to expose their arguments, test their reasons and improve their argumentation ability.

D. LEARNING COMPANIONS AND PEER SIMULATIONS

Recent studies have demonstrated the potential of agentic AI not only to be tutors or graders but peers and learning companions that facilitate social and cognitive learning. Nguyen et al. [38] explored this potential by designing embodied generative AI (GenAI) agents in mixed reality (MR) learning environments for post-secondary education. Their objective was to support active and self-regulated learning and increase human-AI shared regulation. As part of an example of Design Science Research, 26 adult learners participated in embodied agentic AI system-supported MR

learning experiences. In the study, it was illustrated how such agents could improve learner engagement and reduce cognitive load, thereby contributing to developing AI tools to support more agentic and interactive learning.

Complementing this research, Mushtaq et al. [28] investigated the use of Multi-Agent Large Language Models (LLMs) for senior design projects in computer science and engineering. Agents were trained to emulate human collaborative teams that engaged in rich discussions to co-design capstone project proposals. The findings revealed that Multi-Agent LLMs generated more diverse and dynamic problem-solving contexts than single-agent systems. This simulation of real working partnership well resembled faculty expectations and pedagogical goals, better preparing students for future professional partnership in engineering and other technical fields.

Together, these studies demonstrate the ways agentic AI can be constructed to be learning companions—participating in collaborative scenarios, modeling good learning practices and promoting social-emotional learning.

Example: A virtual peer who assists a student in figuring out problems collaboratively, models good help-seeking behaviors and expresses emotional cues such as frustration or excitement to make challenges acceptable and develop resilience within the learning process.

E. TEACHING ASSISTANTS AND CLASSROOM SUPPORT

Jiang et al. [39] introduced the concept of Knowledge Components (KCs) as foundational units of learning within educational systems. Their study emphasized the importance of accurately mapping the relationships among these components to better diagnose and address student learning challenges. To this end, the authors proposed MAS-KCL, a knowledge component graph learning algorithm that leverages a multi-agent system powered by large language models. By incorporating a bidirectional feedback mechanism, the system dynamically assessed edge strengths within the KC graph, thereby refining learning paths and improving instructional efficiency. Experimental validation on both synthetic and real-world educational datasets demonstrated the algorithm's effectiveness in recognizing optimal learning sequences. The approach holds promise for helping educators design more personalized and effective learning interventions, aligning with broader goals of sustainable educational development.

Agentic AI can act as intelligent teaching assistants, supporting instructors in workflow management, analytics and feedback orchestration. Kamalov et al. [27] (field study, $N = 10$ faculty across two institutions) observed a 25 % reduction in administrative load and improved LMS data utilization. Mushtaq et al. [28] and Wei et al. [29] (multi-site trials, $N = 18$ educators) reported enhanced classroom monitoring efficiency and more timely student interventions. These results substantiate high-TRL (6–7) applications in classroom orchestration, linking automation gains directly to measurable teacher time savings (Table 3).

These advancements exemplify how agentic AI systems can serve as intelligent teaching assistants—supporting educators in both instructional planning and classroom management. Rather than replacing teachers, these systems augment their capacity by handling routine yet time-consuming tasks, such as responding to frequently asked questions, summarizing class discussions, monitoring student progress and recommending tailored instructional resources. By doing so, agentic AI can free up educators to focus on higher-order teaching responsibilities, such as mentoring, scaffolding and curriculum design.

Example: An AI-powered classroom assistant that monitors online discussion forums, identifies students who may be disengaged or at risk and provides instructors with daily summaries of participation trends and suggested interventions.

F. CURRICULUM DESIGN AND CONTENT GENERATION

The integration of generative AI into curricula and content design is opening up new possibilities for enhancing instruction and student learning. Tate et al. [40] investigated this potential by aiming the application of generative AI at writing-intensive STEM courses. They developed a writing tool facilitated by AI to guide students through the process of writing and conducted a research study to determine its impact. Building on earlier research on classroom use of digital writing programs, the authors proposed a robust framework for effective integration of generative AI. The findings underscored the importance of enabling students to critically use AI by challenging them to understand, access, prompt, vet and integrate AI-produced content critically. The study also provided best practices for instructors, including setting clear learning goals, defining AI's role in the writing process, iterating curriculum design and reflecting on implementation outcomes.

Complementing this approach, the AIfademic system introduced by Ma et al. [32] demonstrated how agentic AI and Agile methodologies can be combined to support curriculum delivery in complex technical subjects such as computer networks. AIfademic utilized big language models to simulate an elite team of cooperating AI agents, featuring various roles—such as instructors and students—are established via prompt engineering to replicate actual-classroom conversations. The agents co-drafted and delivered content in continuous feedback cycles according to Agile principles of development. A pilot study with instruction in computer networks' transport layer achieved greater student engagement and comprehension, highlighting the system's ability to transform active learning experiences in STEM courses.

Together, these papers demonstrate how top-level agentic systems can be an important facilitator for curriculum design and content generation. Such systems can aid instructors by generating differentiated instruction material, joint lesson plan development, test creation and simulating student behavior to anticipated assignments.

Example: An AI partner that reviews current student performance data and suggests differentiated math exercises for various ability levels, consistent with the instruction objectives a teacher has established for an upcoming lesson.

G. SIMULATION-BASED LEARNING AND VIRTUAL LABS

Simulation-based learning environments are incorporating agentic AI as well to generate immersive, interactive and adaptive learning experiences. Barra et al. [31] demonstrated an AI-driven workflow to facilitate the process of developing healthcare simulation scenarios without sacrificing pedagogical fidelity. The system employed a collection of specialized AI agents to manage key sub-tasks, including learning objective design, patient narrative creation, diagnostic data creation and debriefing content generation. Through a conversational interface, the tool allowed health educators—regardless of technical background—to successfully create high-quality, guideline-compliant scenarios for different clinical environments and learner levels. While the system significantly reduced development time and enhanced ease of use, the authors noted the need for human review to address potential bias and ensure content accuracy. Key lessons learned involved the importance of prompt fine-tuning, co-design and making AI tools user-friendly in medical education.

Extending the idea of agentic AI in medical education, Wei et al. [29] presented MEDCO (Medical Education COpilots), a multi-agent system for the simulation of realistic medical learning environments. MEDCO featured three interactive agents—an AI patient, physician and radiologist—that supported clinical reasoning, collaborative problem-solving and inquiry-based learning. The system aimed to encourage question-asking skills and peer interaction, simulating the dynamics of real-world medical education. Experimental results demonstrated that students educated with MEDCO exhibited more human-like learning tendencies, improved performance and increased interaction with the material, indicating the potential of AI-driven simulation to transform medical education.

Together, these studies illustrate how agentic AI can be used to enhance simulation-based learning in medicine, science and engineering domains. Through the control of scenario flow, real-time feedback and evaluation of learner decisions, agentic AI enables more realistic, dynamic and personalized virtual learning experiences.

Example: A clinical reasoning simulator in which an AI patient dynamically responds to the learner's diagnostic and treatment decisions, poses follow-up questions and challenges the learner's assumptions to encourage deeper insight.

H. SUPPORT FOR LIFELONG AND INFORMAL LEARNING

In addition to formal education, agentic AI has the capability of enabling independent and lifelong learning through enabling learners to develop personalized learning plans, establish independent goals and react to reflective prompts

that instill metacognition and self-regulation [30], [41]. Apart from structuring individual study, the systems can also evolve through time to address evolving learner needs, enabling a continuous cycle of independent development and adaptive learning.

Agentic coaching systems foster self-regulated learning by prompting planning, monitoring and reflection. Controlled evaluations show positive effects on metacognitive strategy use and persistence (Zimmerman [11]; Panadero [15] — randomized controlled trial, $N = 86$ K-12 students, significant SRL-strategy gain, $p < 0.05$). Follow-up field observations indicate sustained learner engagement when adaptive prompts are paired with gradual scaffolding removal. These findings reflect moderate-TRL (≈ 4) systems within the evidence map (Table 3).

Example: A personal learning agent that helps adult learners detect learning gaps, recommend microlearning modules and schedule revision activities on their own.

V. TECHNOLOGICAL FOUNDATIONS

The emergence of agentic AI systems in learning has been made possible by the fast pace of innovation in underpinning AI technologies. Such AI systems often combine language understanding, planning, memory and multimodal interaction abilities to operate autonomously and responsively within complex learning settings. This section outlines the key technological building blocks underpinning agentic AI in learning settings. Figure 4 summarizes the technological foundations of agentic AI in education.

A. LARGE LANGUAGE MODELS AND FOUNDATION MODELS

At the core of many contemporary agentic AI systems are large language models (LLMs) such as GPT-4, PaLM, Claude and other foundation models trained on massive corpora of text data [42]. They offer unprecedented capacity for natural language understanding and generation to enable agentic systems to engage with learners in context-sensitive and fluent manners [43]. Their language flexibility allows teaching agents to comprehend diversified learner inputs, form intricate explanations, pose scaffolding or reflective questions and mimic pedagogical activities such as Socratic questioning, adaptive responses or individualized motivation [44].

LLMs can support a wide variety of educational activities, including content generation, question-answering, summarization, translation, inferencing and story generation [45]. Their contextual sense enables them to tailor explanations to the learners' response, have multi-turn tutoring dialogue and even assess cognitive or affective signals when combined with additional input modalities [46]. These capabilities provide robust grounding for agentic action in teaching, mentoring and collaboration settings.

However, despite being great at what they can do, LLMs have important limitations that inhibit their agentic potential. Number one on the list are their tendency to produce

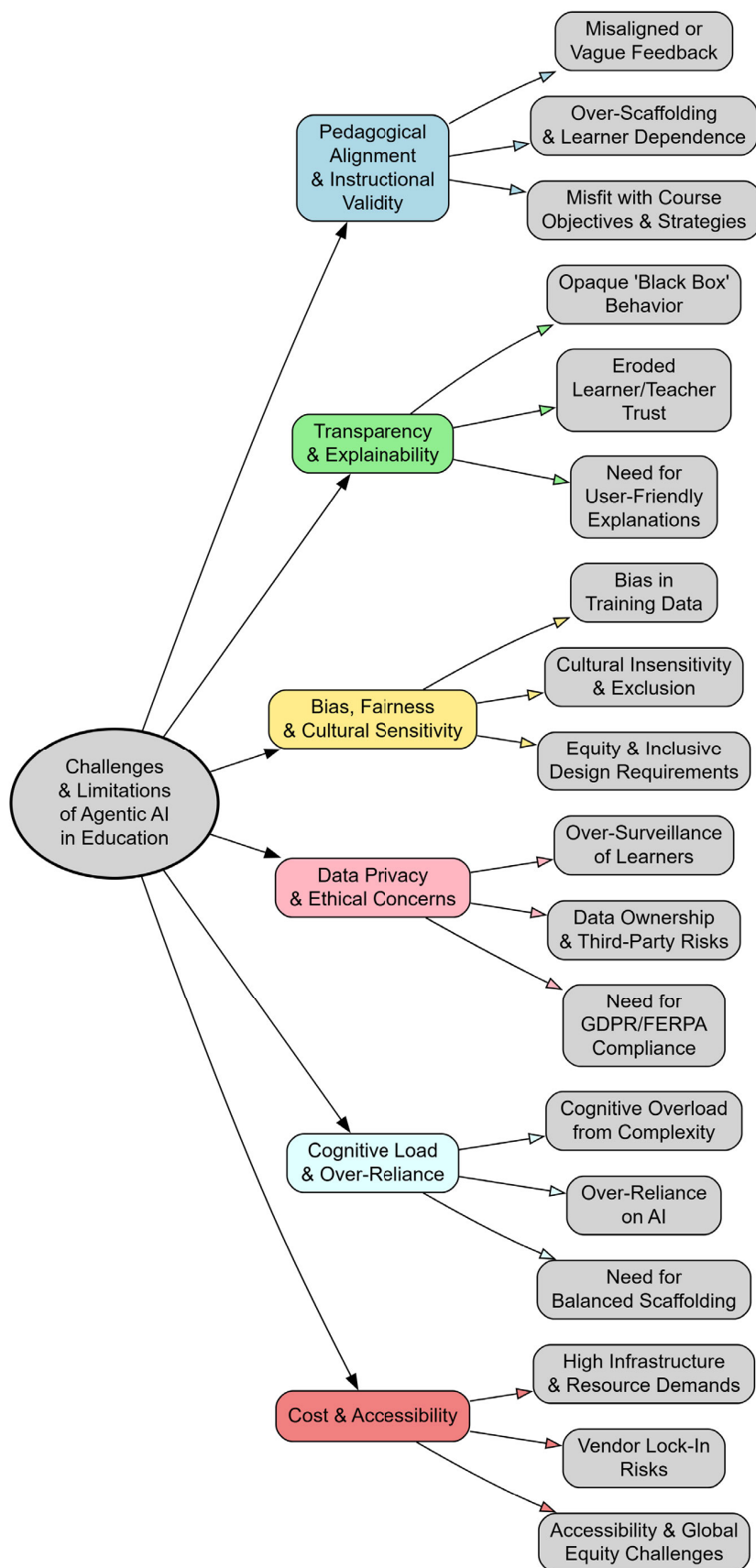


FIGURE 5. Challenges and limitations of agentic AI in education.

unsubstantiated or false claims—commonly referred to as hallucinations—particularly when reasoning about new or ambiguous topics [47]. Additionally, present-day LLMs often lack session-long memory and are heavily limited in terms of being able to store and recall long-term learner information, save for with added external memory modules. Their multimodal understanding capability in real-time is also constrained, with the majority of models being predominantly text-optimized, though this is rapidly changing with the emergence of multimodal foundation models [48].

Foundation models are now stretching their scope from unimodal language-centric tasks to multimodal reasoning by bringing together language and vision, audio and other modalities. Vision-language models (e.g., Flamingo, GPT-4V, Gemini) and multimodal transformers allow the agent to read and respond to images, diagrams or visual comments—a very valuable skill in STEM education and diagrammatic reasoning [49]. These models are also increasingly fine-tuned on educational data sets or aligned to learning targets so that they become more suitable for domain-specific learning environments [50].

B. AGENT ARCHITECTURES

Agentic AI systems are distinguished from standard language models by the inclusion of architectural components that support autonomy, reasoning, memory and interaction with external tools or worlds. One common architecture is the planner–executor model, where the agent decomposes a user-defined goal into a list of subgoals and executes them either internally or by invoking external procedures. The planning component typically maps high-level goals to actual steps and the execution component—often realized on top of an LLM—executes these steps by language generation, API invocation or use of tools. This trend allows for automation of challenging tasks such as multi-step solving or customized delivery of curricula.

Reflective agents extend this capability with intrinsic self-monitoring ability that evaluates the effectiveness of their own actions and modifies approaches in light of feedback. Architectures such as ReAct (Reasoning and Acting) [51] and Reflexion [52] exemplify this functionality, enabling agents to alternate between action and reflection cycles. In educational settings, reflective architectures help agents determine ineffective explanations, modify instructional tactics and stay in sync with learner requirements over time.

Another important class is the tool-augmented agents, connecting LLMs with off-the-shelf software environments such as search engines, calculators, knowledge bases, code execution environments or simulations [53], [54]. These bridges greatly enlarge the agent’s capability beyond the production of pure language, adding data retrieval, numerical computation, visualization and interactive experimentation [55]. For example, an educational tool-augmented agent can combine natural language explanation and graph plotting or algebraic manipulation in real time.

Finally, multi-agent systems define a distributed architectural pattern in which individual agents with specific roles, abilities or pedagogies collaborate on complex learning tasks [56]. Agents are able to simulate peer debate, imitate teacher–student interaction or divide workload across tutoring, mentoring and administrative functions [57]. Such architectures are specifically relevant in group learning environments, project-based settings or simulations in which diversity of opinion and dynamic engagement are desirable [58].

C. MEMORY AND CONTEXT MANAGEMENT

Continuous, personalized and flexible interactions in agentic AI systems are largely dependent upon robust memory and context management capabilities. Unlike classical chatbots or short-context models, agentic systems need to be capable of establishing consistency across longer interactions—over task sequences, sessions and learning cycles—calling for memory architectures beyond the ephemeral context windows of large language models (LLMs) [59]. To meet this need, such systems tend to integrate short- and long-term memory processes that play complementary roles in guaranteeing relevance and personalization.

Short-term memory defines the direct dialogue history saved in the model’s prompt or session frame [60]. It allows for the agent to respond fluently within a single interaction, maintaining discourse coherence and local context. But due to token capacity and potential decay in relevance with time, such memory must be sanitized or pruned repeatedly. Long-term memory rather stores interaction histories, learner profiles, knowledge states and previous performance data in external storage such as vector databases, semantic memory graphs or structured knowledge bases [61]. These storage facilities are accessed dynamically by semantic retrieval mechanisms, thus enabling agents to retrieve relevant information, recall previous sessions and adjust behavior in accordance with learner information accumulated.

Redundancy management and information overload are among the fundamental memory integration difficulties. Context distillation techniques overcome this through summarizing, abstracting or selectively filtering prior interactions [62]. Such techniques ensure that the model is only presented with pedagogically important and timely information, promoting efficiency while allowing richness of context. For example, an agent may summarize a week’s worth of tutorial conversation into a top-level learning progress report or extract prominent misunderstandings for remediation planning.

D. PLANNING AND REASONING

Planning and reasoning are central capabilities of agentic AI systems, enabling them to move from reactive reactions towards intended, goal-oriented action. In the educational environment, where learning tasks often last through multiple steps—diagnosing a misconception, leading one through an exercise and looking at outcomes—planning is essential

TABLE 4. Responsible deployment guidelines for agentic AI in education.

Scaffolding & Fading. Configure instructional support to decrease as learner autonomy increases; monitor guard-metrics such as autonomy ratio (independent actions / total actions), help-seeking balance, and offloading index.

Help-Dependence Monitoring. Track frequency and duration of automated assistance requests; flag excessive reliance when help requests exceed 30% of total learner actions; embed reflective prompts to restore self-regulation.

Privacy & Data Governance. Apply data minimization, explicit purpose limitation, and role-based access; enforce retention windows and maintain verifiable audit trails.

Bias Audits & Datasheets. Conduct pre-deployment and periodic disparate-impact assessments; document dataset and model provenance through standardized datasheets; retain decision logs for transparency.

Decision Traceability Logs. Record agentic decisions, triggers, and outcomes with timestamps to enable post-hoc review, accountability, and pedagogical explainability.

Interoperability & Evaluation. Integrate via LTI, Caliper, or xAPI standards; ensure traceability for learning analytics pipelines and reproducible field evaluations.

to guarantee coherence and pedagogical suitability [63]. Agentic systems rely on various mechanisms in order to do this, with both statistical and symbolic strategies.

One of the top approaches in large language model (LLM)-based agents is chain-of-thought (CoT) prompting, which produces middle-level steps of reasoning in natural language. CoT prompting assists agents in decomposing challenging problems into computationally tractable subgoals, improving explainability as well as the quality of the explanations provided to students [64]. CoT prompting has been particularly valuable in logical reasoning, multi-step computation or justification-based answer tasks, assisting in simulating the pedagogical practice of scaffolding through the guided process of thinking [65].

Besides implicit reasoning, explicit programmatic planning is also integrated into most agentic systems. This involves the use of symbolic planners, decision trees or hierarchical task decomposition structures to represent action sequences in time. By serializing step-by-step dependencies and checking potential outcomes, agents can execute coherent plans—such as creating tailored learning pathways or supporting peer-to-peer collaboration activity—while dynamically adapting based on learner progress.

More advanced agentic action involves meta-reasoning, where agents recognize and deliberate about their decision-making themselves [66]. Meta-reasoning allows agents to

adjust strategies given feedback, unexpected learner action or changes in task objectives. For example, when an instructional approach fails to elicit the anticipated learner reaction, a meta-reasoning agent may spot the failure, change its instructional plan appropriately and restart with a different approach.

E. MULTIMODAL INTERFACES AND EMBODIMENT

Agentic AI systems are being designed to operate through multimodal interfaces that support diverse inputs and outputs, enabling more natural, inclusive and contextualized interaction. They can recognize and interpret spoken words, text input, hand gestures, facial expressions, handwritten input and visual signals and adaptively respond through synthesized speech, animation, graphical output or motion. Such multimodal interaction not only enhances accessibility for diverse learners with varying needs and preferences, but also more accurately mirrors the way humans communicate and learn through sensory modalities [67].

Embodiment—whether in the form of animated avatars, virtual agents in augmented or virtual reality (AR/VR) environments or physically embodied social robots—is essential in expressing social presence, affective connection and interpersonal relationships in digital learning environments. Evidence shows that embodied agents can facilitate learner motivation, establish trust and provide more sustained interaction, especially in collaborative or affective learning contexts [68]. Perceived presence, personality and responsiveness of the agent greatly improve the experience and performance of the learner, above all in contexts of empathy, encouragement or high-grained feedback.

The underlying facilitating technologies are changing rapidly. Speech-to-text and text-to-speech streams provide real-time dialogue interfaces, whereas computer vision and sensor fusion techniques enable gesture recognition, affect detection and spatial presence. Real-time animation engines and motion capture enable avatars to express human-like behaviors, which contributes to the experience of intentionality and rapport. With these modalities fused together, agentic AI systems end up having the capability to identify advanced learner states and respond in socially and pedagogically effective ways [69].

F. INTEGRATION WITH LEARNING PLATFORMS

If agentic AI is going to be effectively implemented in practice in education, it must integrate with existing digital ecosystems, such as learning management systems (LMS), content-delivery platforms, assessment tools and institutional data infrastructures. This integration is typically achieved by conformity mechanisms and interoperability contexts such as Learning Tools Interoperability (LTI) and application programming interfaces (APIs) that allow agentic systems to reuse course materials, learners' accounts, gradebooks and live feeds of activity [70]. Interconnection in this manner allows agents to interact frictionlessly with learners and

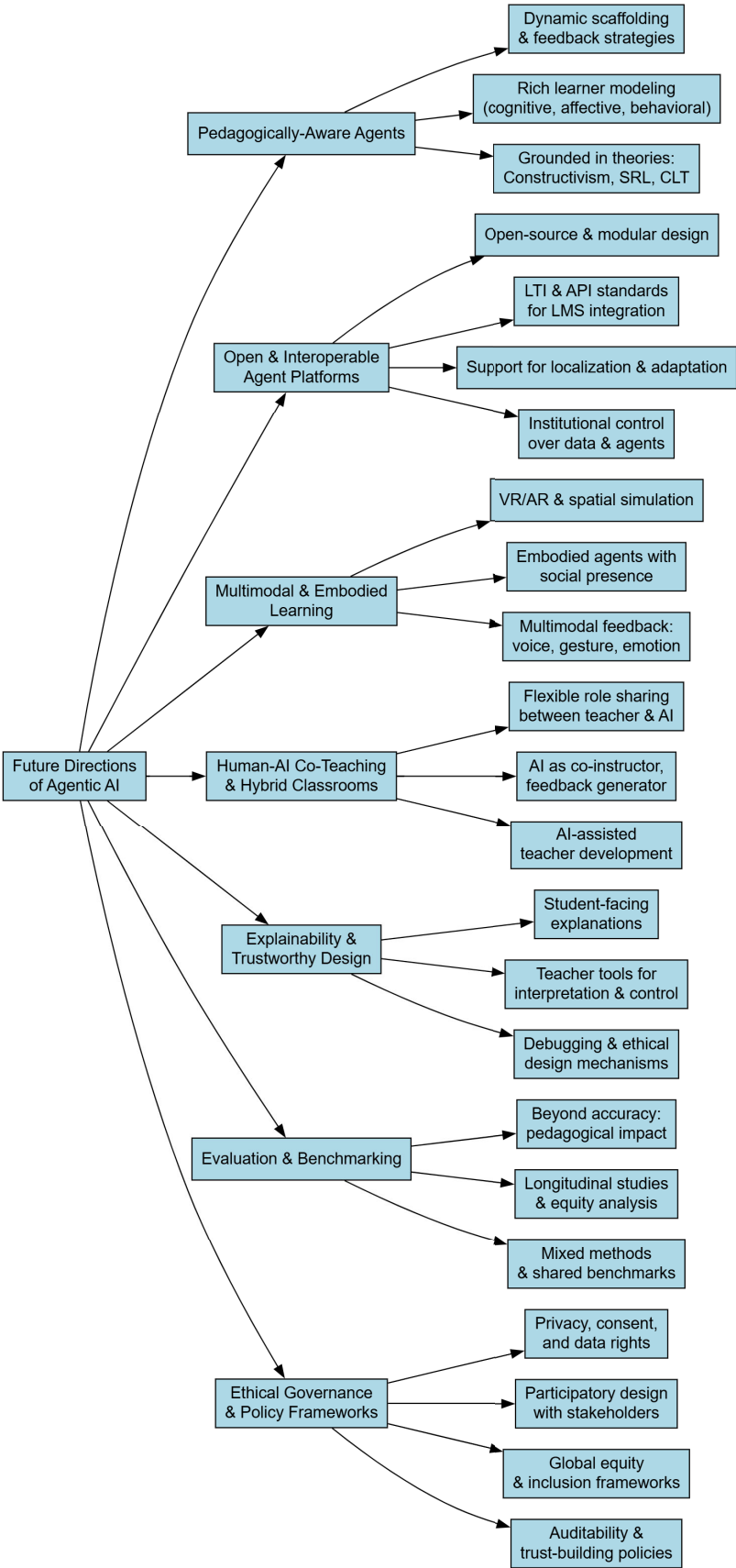


FIGURE 6. Future directions for agentic AI in education.

instructors using familiar contexts while benefiting from institutional infrastructure.

An end-to-end data infrastructure is similarly necessary to support agentic conduct. Agentic AI infrastructure relies on the real-time gathering, processing and analysis of interaction data—e.g., navigation pathways, engagement rates and performance measures—to build dynamic learner models and adjust in real time. Data pipelines for these applications must be able to track activity over many tools and environments so that agents can create a full picture of a learner's path, requirements and preferences. To facilitate this, integration with learning analytics dashboards can provide actionable feedback both for students and teachers that can further improve the agent's ability to scaffold learning and recommend interventions [71].

Equally important is the exercise of secure privacy, access control and data governance frameworks [72]. As the data in education is sensitive and governed by guidelines such as GDPR and FERPA, agentic systems must incorporate mechanisms of informed consent, role-based access and secure data handling. Ethical deployment also necessitates transparency in how the agentic decisions are being taken, particularly when affecting learner pathways or assessments. As such systems become increasingly autonomous, it is not merely a technical need but a pedagogical and ethical requirement to preserve human oversight and institutional responsibility through clearly articulated integration protocols.

VI. BENEFITS AND OPPORTUNITIES

The study [73] explored predictors of sustained use of AI facilitators in higher education settings with a focus on self-regulation, academic buoyancy, trust and agentic engagement. Data was collected from Omani students and findings were that self-regulation and academic buoyancy positively affected the perceived ease of use and usefulness of AI facilitators, which impacted agentic engagement and trust and thus led to continuance usage of AI facilitators. Academic buoyancy was the mediator between self-regulation and continuance usage. The study provided implications for theory and practice.

A. PERSONALIZED AND ADAPTIVE LEARNING AT SCALE

Agentic AI holds a revolutionary promise to deliver highly differentiated learning experiences to very large and diverse groups of learners. Monitoring learner activity, interest and performance in real-time, such systems are capable of dynamically adapting pedagogical techniques to meet the individual needs of each learner. This agency enables agents to vary content difficulty and rate to accommodate learners' current skill levels, so challenges are both engaging and appropriately scaffolded. Moreover, agentic systems can provide timely, context-sensitive feedback and explanations that are relevant to the learner's individual misunderstandings or misconceptions, thereby promoting enhanced understanding and retention [74].

Apart from adapting content presentation, agentic AI supports a wide variety of learning styles and disability requirements, including education. Whether students learn most effectively through visual, auditory or kinesthetic channels or must be accommodated for disabilities, adaptive agents can modify their channels of interaction and modes of teaching in response. In addition, such systems support students in monitoring their own learning through metacognitive resources that invite reflection, goal-setting and self-regulation [75]. By enabling learner autonomy and awareness, agentic AI enables the development of lifelong learning capabilities that extend far beyond the confines of individual learning contexts.

The scalability of agentic AI systems provides the capability to transfer these personalized and adaptive strengths to truly massive numbers of learners simultaneously, which corrects long-standing educational access and equity problems. These systems can close achievement gaps by providing personalized assistance to disadvantaged or underprivileged groups, particularly in organized classroom settings but also in unorganized or distance learning situations [76].

B. AUGMENTATION OF EDUCATOR CAPABILITIES

Agentic AI systems are not intended to substitute for teachers but rather act as intelligent assistants that minimize administrative burdens and increase teaching efficiency. By automating routine, time-consuming activities such as marking open-ended exercises, the systems will provide prompt, consistent and rich feedback, freeing teachers to concentrate on individual student support and pedagogy. Secondly, agentic AI is able to monitor the learning and engagement patterns of students in real-time, generating alerts whenever signs of disengagement or misconceptions are sensed [77]. Anticipatory insight allows teachers to intervene promptly and adjust teaching to address unique student requirements.

Drawing on learner analytics, agentic AI is able to offer context-aware recommendations for differentiated instruction, allowing teachers to design instructional experiences that address various levels of skills and learning styles. Such proposals can help teachers design customized lesson plans, formative assessments and grading criteria according to curriculum goals and learner profiles [78]. These platforms can also help teachers effectively produce curricular content and testing, reducing preparation work to a significant level while ensuring content quality and conformity with teaching standards.

C. ENHANCED ENGAGEMENT AND MOTIVATION

Agentic AI holds enormous promise for encouraging increased learner engagement by capitalizing on interactive, conversational and game-like experience that capture learners' attention and sustain motivation. By simulating social presence through individualized conversation and response adaptation, these systems can create a sense of

belongingness and cooperation, which is generally absent in online or self-paced learning environments [79]. Personalization enables the agent to tailor its interaction based on the learner's interest, learning trajectory and affective state with immediate encouragement and feedback that supports persistence in spite of hurdles.

Furthermore, agentic AI can also include game-like elements such as reward, challenge and narrative development, making learning activities game-like experiences that trigger curiosity and intrinsic motivation. This responsive approach not only enhances cognitive commitment but also facilitates affective commitment, rendering learners more active and confident in the learning process [80].

For instance, a history student may have had role-played dialogue with AI characters that embody historical actors so that the student could explore alternative perspectives, ask questions and practice critical thinking within an interactive, situated environment. Embodied interaction in this way makes distal or abstract content more accessible and more memorable and enables active learning. These experiences reveal the potential for agentic AI to transcend traditional pedagogy by facilitating emotionally rich and socially rich learning environments that foster long-term engagement and motivation.

D. SUPPORT FOR LIFELONG AND INFORMAL LEARNING

Agentic AI systems extend the reach of learning way past the confines of standard classrooms, reaffirming learners throughout their lives in many informal and professional settings. These smart personal assistants offer personalized direction that adapts in real-time to individual goals, interests and evolving skill sets [81], making lifelong learning convenient and enjoyable. By tracking skill accumulation of users over time, agentic systems can dynamically restate learning objectives and recommend personalized paths that align with personal goals and professional trajectories.

Besides, the agents can provide relevant resources autonomously, i.e., courses, articles or practice exercises, chosen from various repositories and sites in order to aid in just-in-time learning. The ability to seamlessly integrate with individual and professional development applications, i.e., digital portfolios, certification systems and in-workplace learning management systems, enables full-fledged enablement of skill acquisition and career development [82].

E. NEW PEDAGOGICAL PARADIGMS

The development of intelligent, interactive agents challenges the very pillars of instructional design and educational theory, offering scope for new pedagogical approaches to fully exploit the capabilities of agentic AI. For instance, AI systems can enable learning through inquiry by connecting learners to Socratic questioning, encouraging more reflection, critical thinking and self-directed exploration [83]. Such discourse-based interactions foster a more active and student-oriented learning environment where the learner is prompted to construct knowledge together with the agent.

Second, AI agents can be programmed to perform as peer learners or collaborators and create enriched peer learning environments that surpass human capabilities. Such virtual peers can participate in discussions, present varying perspectives and offer scaffolding of social learning processes, thus enabling collaboration and co-construction of meaning in physical and online classrooms.

Moreover, simulations based on AI offer scope for learning by doing in challenging, high-risk subjects such as medical diagnosis, ethical decision-making and engineering design. By putting students in authentic environments where they can try things out, receive immediate feedback and explore consequences in safety, these simulations enable active learning and the development of transferable skills.

F. DATA-INFORMED DECISION MAKING

Agentic AI systems produce huge and high-granularity student interaction, engagement behavior and performance data that offer unparalleled insights to support decisions at all levels of education. Classroom level data supports real-time learning analytics improving formative assessment in offering teachers timely information about student learning, misconceptions and improvement [33]. Timely feedback allows dynamic instructional adaptation based on individual and group needs, resulting in more responsive and effective teaching.

Outside of individual classrooms, grouped learning data can inform curriculum design and enhancement through information regarding what content, activities or sequences yield maximum learning gains. Data-driven curriculum design ensures that instructional materials are up-to-date, effective and relevant to learner needs.

At the institutional and system levels, analytics from agentic systems can unveil trends that indicate systemic inequities or persistent learning barriers. With those unveiled, educational policymakers and leaders can best target interventions and allocate resources in areas that most require them.

G. GLOBAL REACH AND ACCESSIBILITY

Agentic AI systems hold remarkable promise for expanding the reach of quality education across diverse geographical, socioeconomic and linguistic contexts. By operating continuously without the constraints of traditional schooling hours, these systems can provide personalized learning support around the clock, making education more flexible and accessible for learners in different time zones or with irregular schedules [84]. Their capacity to process several languages and dialects brings inclusivity a step further as it allows students to learn either in their own or chosen language, which is a key to understanding and motivation.

In addition, agentic AI can be designed to be extremely flexible across a wide range of devices, ranging from smartphones to low-end tablets and function optimally under varied bandwidth conditions, overcoming typical technological obstacles faced in low-resource and rural environments.

Its flexibility enables learners with restricted internet access or old machinery to be integrated into the benefits of leading-edge learning technologies.

As one tangible example, AI-powered tutors on a mobile platform can provide personalized, real-time guidance to students in remote or underprivileged regions, supplementing or even substituting finite human teaching resources. Such platforms can ensure round-the-clock practice, instant feedback and scaffolding that otherwise might be impossible, enabling bridging of gaps in learning and democratization of schooling opportunities everywhere [85].

VII. CHALLENGES, PEDAGOGICAL RISKS, AND MITIGATION PRACTICES

This section translates pedagogical risks into actionable mitigation strategies, emphasizing measurable guardrails for scaffolding, traceability and ethical oversight. While having promising promises, agentic AI systems in education face fundamental challenges that restrict their reliability, equity and pedagogical appropriateness. Their boundaries must be critically encountered to achieve the potential of such systems for improving rather than degrading human-centered education. Figure 5 summarizes the challenges and limitations of agentic AI in education.

A. PEDAGOGICAL ALIGNMENT AND INSTRUCTIONAL VALIDITY

While agentic AI systems have much to offer to enhance educational experience, they need not represent good pedagogical practice. Without deliberate and considerate design, these systems can offer support that is contraindicated for effective learning, for instance, reinforcing shallow engagement, disseminating inaccuracies or instilling unproductive learning habits. Among the most important of these problems are those that arise when the agents provide misaligned feedback with the learning goals—such feedback should be incorrect, unachievable vague or too general and thereby prevent learners' progress rather than facilitate it. Misaligned feedback can confuse students or further solidify mistakes and therefore decrease the learning value of the interaction [34].

Moreover, agentic systems inadvertently have the over-scaffolding effect, where excessive guidance cuts down on possibilities for fruitful struggle or thinking critically. The balance between helping and autonomy is fine; if agents overindulge in providing solutions or cues too soon, students will be over-dependent and not cultivate deeper problem-solving abilities or self-regulation strategies.

Second, agentic AI may not always coherently align with some course objectives, instructor strategies or diverse learner needs. Without alignment with the broader instructional environment, such as curriculum standards and pedagogical models, agents can produce incoherent or disconnected learning experiences that mislead learners and instructors as well [86].

Misalignment of this type can destabilize the integrity of the learning process and render technology-enriched learning interventions ineffective.

Risk: Misaligned feedback and over-scaffolding. Agents may provide guidance inconsistent with course objectives or intervene too frequently, leading to dependency and reduced problem-solving autonomy.

Mitigation: Design for progressive fading and goal-alignment audits. Implement fading protocols—gradually reduce hint frequency after each successful attempt; target a help-seeking ratio (AI-initiated hints / total student actions) below 0.3. Conduct quarterly checks on alignment of content with course faculty to ensure that agent prompts align with learning outcomes. Integrate reflective questions (“Explain your reasoning”) every three automated cues in a way that maintains high metacognitive activity.

B. TRANSPARENCY AND EXPLAINABILITY

Agentic AI systems, particularly those grounded in large language models and complex decision-making architectures, have a tendency to function as opaque “non-interpretable systems,” obscuring the driving logic underlying their actions and outputs. This lack of transparency presents special challenges to various stakeholders in educational environments. For learners, a lack of explainability in feedback can lead to confusion or suspicion about why an agent makes specific suggestions, feedback or recommendations, which can undermine learner engagement and agency. Without a window into the agent's reasoning process, learners can find it hard to critically evaluate or respond effectively to support provided [87].

Teachers and ultimately instructional quality and learning outcomes, might be unable to trust agentic AI suggestions without understandable explanations. This lack of transparency can hinder teachers from being able to effectively integrate AI-derived insights into pedagogical decision-making or intervene when the system's advice conflicts with domain knowledge or contextual requirements [88]. Also, the lack of interpretability hinders the validation and improvement of agentic systems from a research perspective since it diminishes the possibility of error diagnosis, biases or unwanted behavior.

Risk: Opaque agent reasoning and low trust. Students and teachers will not comprehend why the system gives certain recommendations, which will lose credibility and interpretability.

Mitigation: Enable explainable feedback and decision traceability logs. Offer a “Why this suggestion?” alternative that reveals brief rule-based or probabilistic reasoning. Maintain decision traceability logs—timestamped agent rationale, data sources, and decisions—available for instructional auditing. Train users to interpret confidence indicators (e.g., color-coded uncertainty bars) during feedback review sessions.

C. BIAS, FAIRNESS, AND CULTURAL SENSITIVITY

Agentic AI systems, particularly those founded on large-scale language and multimodal models, are susceptible to replicating and even amplifying biases in the training data. These biases, be they gender, racial, socioeconomic, linguistic or cultural, can result in unfair treatment of learners, influencing the quality, appropriateness and relevance of their learning experience. For instance, a computer tutor might react differentially to inquiries posed in non-standardized dialects or the examples, metaphors or values it selects might unknowingly promote culturally mainstream perspectives [89]. These divergences might lead to alienation, lowered learner motivation or the reinforcement of harmful stereotypes.

Aside from overt discrimination, even more subtle forms of cultural insensitivity—such as the erasure or manipulation of minority histories, traditions or worldviews—can subvert the inclusivity of online learning spaces [90]. Even user-behavioral personalization through algorithms can be inequitable across demographically diverse groups if training data is inadequate in diversity or system design ignores intersectionality of experience.

Risk: Embedded data or linguistic bias. Training data can overrepresent certain cultures or demographics, which can cause inequitable or culturally insensitive outputs.

Mitigation: Conduct fairness audits and inclusive co-design. Conduct pre-deployment disparate-impact testing on at least three demographic groups (gender, language, geography). Recruit diverse teachers in rotating through prompt and dataset evaluation rounds. Assume model datasheets detailing dataset construction, indicators of bias, and mitigation approaches. Retrain flagged violations of fairness beyond $\pm 10\%$ performance difference.

D. DATA PRIVACY AND ETHICAL CONCERNS

The use of agentic AI in education typically relies on access to rich streams of learner data, including interaction logs, behavioral cues, performance data and even biometric data. While such data can enable more personalization and responsiveness, it also engenders grave ethical and legal issues regarding privacy, consent and data stewardship. For learners—particularly minors and members of vulnerable populations—the risks of over-surveillance and digital monitoring are especially urgent [91]. Continuous data collection may lead to environments where learners feel scrutinized, potentially inhibiting authentic engagement and expression.

Moreover, questions of data ownership, storage, access by whom and for what purposes remain unresolved in most environments. Lack of visibility of data flows and reliance on third-party providers further amplify the potential for abuse, e.g., profiling, discriminatory actions or unauthorized release of sensitive information [92]. These risks are further elevated in cross-border applications where legal protections may vary or even differ.

Risk: Over-monitoring students and fuzzy data ownership. Ongoing data logging can be a violation of privacy rights or institutional policies (e.g., GDPR, FERPA).

Mitigation: Apply privacy-by-design principles. Collect pedagogically useful data alone and obfuscate identifiers when you ingest them. Set up retention windows (<12 months) and access permissions that are based on role. Give students dashboard controls such as viewing, exporting, or deleting their data. Keep independent audit trails distinct for all data exports or external API requests.

E. COGNITIVE LOAD AND OVER-RELIANCE

While learning-supportive agentic AI systems are developed, their implementation in instructional environments can actually increase cognitive load or encourage dependency. When students are required to interpret agent complexity, structure interactions or reconcile system inconsistency, cognitive resources may be reallocated away from the primary learning task. This additional layer of interaction complexity may be particularly onerous for struggling students or those with weak digital literacy. Also, over-supportive agents—especially those too hasty to provide pre-packaged solutions or too prescriptive in their comments—may lead learners to over-rely, reducing productive struggle, self-generated questioning and the development of critical thinking abilities [93].

In such cases, the learner will capitulate too easily to the agent, bypassing processes of reflection or problem-solving, both undermining autonomy and long-term competence. This process has the potential to convert learners into passive recipients rather than active constructors of knowledge. In order to offset such influence, agentic systems need to be designed with attention to balanced scaffolding—effortful involvement, metacognitive monitoring and fading assistance as the learner moves toward expertise [94]. Thoughtful management of human and AI inputs, informed by cognitive load theory and self-regulated learning principles, is required to offer agentic AI that enhances, instead of hindering, the learner's cognitive and motivational growth.

Risk: Cognitive overload and learned helplessness. Complex interface or over-supportive commentary can distract or encourage passivity.

Mitigation: Simplify interaction and balance assistance levels. Set agent prompt per task limit at one. Keep track of the autonomy ratio (independent moves / all moves) and have ≥ 0.7 across a learning unit. Implement regular “reflection checkpoints” asking students to summarize rationale before providing further assistance. Employ adaptive fading curves whose likelihood of intervention decreases as the learner's proficiency increases.

F. COST AND ACCESSIBILITY

Although they have the potential to transform learning environments, the use of highly sophisticated agentic AI systems is commonly restricted by huge financial, infrastructural and institutional hurdles. Developing and maintaining such systems typically requires colossal investment in computer

resources, expertise and software integration that might not be readily available in low-resource learning contexts or in smaller institutions. In addition, reliance upon proprietary technologies and commercial platforms has vendor lock-in risks, shortening the ability of schools and universities to audit, adapt or manage their systems based on local pedagogical values, data policy or cultural norms.

Risk: High cost in infrastructures and unequal access.

Advanced agentic systems can be reliant on costly cloud services, making low-resources institutions ineligible.

Mitigation: Prioritize lightweight, interoperable, and open solutions. Choose open-source or on-device models that can work offline. Make use of LTI/xAPI compatible modular APIs when deploying on existing LMS sites. Benchmark energy and computational cost per learning hour; keep below 0.5 GPU-hr/100 students as sustainability indicator. Encourage common infrastructure consortia to spread the cost of compute and maintenance.

To incorporate high-level ethical and pedagogical principles into practices, Table 4 outlines a compact set of responsible deployment guidelines for education's agentic AI systems. Built-in operational or measurable criteria facilitate implementation and auditability in the real world. Instead of high-level principles, the controls function like guardrails: they define autonomy balance (e.g., through an autonomy ratio), regulate learner dependency, and ensure transparency through data governance, bias audits, and traceability logs. They all work towards a shift from principle-based ethics towards evidence-informed accountability, tracing system behavior back towards pedagogic intent and learner well-being.

VIII. FUTURE DIRECTIONS

As agentic AI technologies mature, several directions for strategic development of their learning capabilities have been explored. These directions in future development vary from technical innovation through pedagogical integration, ethical frameworks and policy formulation. Figure 6 summarizes future directions for Agentic AI in education.

A. TOWARD TRULY PEDAGOGICALLY-AWARE AGENTS

Though contemporary agentic AI systems excel in instructional delivery and dialogue generation, they often lack rich interweaving of pedagogical theory and practice. The majority of recent systems have used surface-level conversation fluency without resorting to grounded ideas of learning processes. To fill this gap, the future of research should focus on how to develop agents that can dynamically model and instantiate essential instructional strategies—such as scaffolding, support fading, formative feedback and metacognitive prompting—into rich learning environments. These strategies should not be expressed as rigid templates but instead instantiated in context-dependent reasoning that adjusts based on the learner's goals, needs and path over time [95].

Pedagogically-intelligent agents must, therefore, also possess the ability to personalize through the use of adapting behavior on the basis of rich learner models. These models must integrate cognitive, behavioral and affective signals to infer learner profiles and states moment by moment, enabling more fine-grained and opportune interventions. The ability to recognize confusion, frustration or disengagement and respond to it is critical in maintaining learner motivation and optimizing learning gains [96].

Most importantly, the design of these agents must be guided by good learning theories—like constructivism, cognitive load theory and self-regulated learning—such that agent actions are founded on empirically established principles rather than ad-hoc heuristics. This vision is to be achieved through the interdisciplinary collaboration between artificial intelligence researchers and learning scientists in the joint development of frameworks for integrating pedagogical intentionality into the agent's architecture, conversational patterns and choice logic.

B. INTEROPERABLE AND OPEN AGENT PLATFORMS

Long-term sustainability and equitable impact of agentic AI in education will depend on the development of interoperable, modular and open platforms. Siloed or proprietary systems risk reinforcing digital divides by constricting access, flexibility and institutional agency. In contrast, open-source systems deliberately designed for educational settings can minimize the barrier to entry for researchers, developers and educators around the world [97]. Such frameworks enable collective innovation, encourage openness in agent action and enable collaborative improvement among institutions.

For complete interoperability, agentic systems must offer well-documented APIs and support standards such as Learning Tools Interoperability (LTI), allowing them to interoperate directly with nearly any learning management system (LMS), content store and data analytics platform. This platform interoperability will render agents viable to execute within existing educational infrastructure, leveraging institutional resources without the cost of overhauls [98].

Moreover, the architecture of reusable, extensible pedagogical agents—self-contained modules that enact instructional strategies, dialogue patterns or assessment strategies—can facilitate teachers and technologists to localize and extend agentic functionality to meet specific curricular, cultural or linguistic needs. Such versatility is especially important in addressing under-endowed educational contexts, where local adaptability is paramount to relevance and impact.

Notably, open and interoperable agent platforms also enable institutional control over infrastructure and data. By enabling the hosting, adaptation and governance of agentic systems by institutions, these platforms minimize dependence on commercial providers and allow for the coordination of AI deployment with local values, privacy requirements and pedagogical issues.

C. MULTIMODAL AND EMBODIED LEARNING EXPERIENCES

The incorporation of multimodal and embodied interaction capability into agentic AI technology is revolutionizing the nature of the interactions of learners with learning content. Speech-to-text recognition, vision and gesture recognition advances now allow agents to operate across modalities and offer richer, more immersive learning settings [99]. One particularly stimulating area is the use of virtual and augmented reality (VR/AR) agents for experiential learning. These agents may lead students through immersive simulations, virtual field trips or scenario-based problem-solving that exceeds the limitations of text or two-dimensional displays. In so doing, they facilitate embodied cognition—where spatial context and physical action are key to understanding and remembering.

Even more revolutionary is the development of embodied conversational agents, such as robot-based instructors or avatar-holding personas in computer simulated worlds. Embodied agents bring social presence, a key factor in sustaining interest and motivation, particularly for younger students [100]. Through facial expressions, body postures or intonation, embodied agents can evoke qualities of human communication commonly absent in traditional digital learning, building trust and rapport between learners and agents.

Moreover, multimodal feedback and assessment mechanisms are expanding the evaluation capabilities of learning AI systems [101]. Instead of relying on written responses or multiple-choice inputs, agentic systems can now assess speech patterns, facial responses, body movement and other cues to gauge learner understanding, emotional state or level of engagement. Such multimodal knowledges support more responsive and adaptive learning pathways, of particular value in early childhood learning, second language acquisition and special needs educational settings where various modes of communication are required.

D. HUMAN-AI CO-TEACHING AND HYBRID CLASSROOMS

Rather than positioning artificial agents as replacements for human instructors, a more practical and morally defensible vision starts with agentic AI systems as cooperative co-partners—co-instructors, guides or intelligent classroom aides. Within the new reality of hybrid classrooms, the division of labor among people and AI is agile and responsive, with functions flowing back and forth according to instructional context, pedagogical purpose and learner needs. For example, agentic systems may assume responsibility for tasks such as personalized practice, immediate feedback or supporting differentiated instruction, while the instructors are responsible for classroom dynamics, engagement relation and second-order pedagogical decisions.

Flexible distribution of tasks dynamically between teachers and AI agents guarantees enhanced instructional effectiveness without compromising human judgment [102]. In such contexts, AI can take on mundane or data-driven responsibilities—such as monitoring progress of students,

identifying misconceptions or recommending customized interventions—leaving instructors to focus on creative, affective and contemplative aspects of teaching. Moreover, agentic systems can also act as facilitators for peer learning by framing collaborative activities, triggering discussion or ensuring equitable participation, enriching the social aspects of learning.

Above all, hybrid classrooms with agentic AI can further enable teacher professional development. Based on the analysis of classroom interaction, learning trajectory and instructional methods, these systems can give reflective feedback to teachers, offering pedagogical recommendations or suggesting alternative approaches [103]. These abilities open up a new type of AI for teachers—those that support educator agency rather than replace it.

Finally, the co-teaching model envisions AI as scaffolding and not replacement—amplifying the human capacity to individualize teaching, collaborate with diversity and respond flexibly to students. When thoughtfully designed and implemented, these blended models can scale good education while preserving the unique value of human teaching.

E. EXPLAINABILITY AND TRUSTWORTHY DESIGN

Wide adoption and meaningful integration of agentic AI systems within education will rely on their ability to foster transparency, accountability and user agency. For students to engage trustfully with these systems, the agents need to provide comprehensible and accessible explanations of their reasoning and decisions. This type of explainability enables students to question, challenge and further understand guidance or feedback presented to them, thereby supporting metacognitive development and facilitating more informed engagement [8]. Beyond student-facing transparency, models configured to explain or justify themselves in pedagogically appropriate ways help bridge the gap between complex AI activity and learning objectives, aligning machine behavior with human instructional intentions.

Equally important is the provision of interfaces and tools whereby teachers can inspect, interpret and individualize the actions of agentic systems [104]. Teachers must play a central role in controlling the appropriateness of agent interventions, their alignment to curricular demands and individualizing agent approaches to the specific needs of their students. By making agent decision-making interpretable and controllable, such tools establish teacher trust and facilitate the responsible deployment of such technologies in classrooms and learning environments.

Second, explainability serves a significant function in the iterative development and debugging of agentic AI, allowing designers and researchers to identify failure modes, biases or unwanted behavior early in the deployment pipeline. This fosters continuous improvement and aligns with ethical AI governance through the renderability of agents to be audited, corrected and aligned with evolving educational values. In all, the incorporation of explainability and trustworthiness in designing agentic AI is at the core of building systems that

not only perform optimally but also gain the trust of learners, educators and institutions.

F. EVALUATION AND BENCHMARKING IN EDUCATIONAL CONTEXTS

As the educational agentic AI models continue to advance and get more complex, so does the urgency for strong, domain-based assessment frameworks. The standard metrics revolving around language performance or general measures of achievement do not capture the full sweep of educational potential these models represent. Future research must develop sound measurement techniques that not only test the accuracy and coherence of generated content but also the pedagogical quality of agentic interaction and their effects on learners' performance. These include assessments of cognitive gain, skill development, motivation and learner involvement in different learning environments.

Longitudinal designs are particularly critical to understand the longer-term effects of agentic AI on learning trajectories, access equity and participation among multiple demographic groups. Longitudinal investigations can reveal whether technologies bridge achievement gaps or unintentionally widen currently prevailing gaps. As a complement to quantitative measures of outcome, mixed-method techniques using qualitative data from teachers, students and other constituents provide rich insights into user experience, trust, perceived utility and contextual determinants of implementation success.

Establishing standard evaluation procedures and shared benchmarking datasets will enable the educational research community to compare agentic systems in a rigorous manner, distinguishing genuinely effective innovations from novelty-catching or cosmetic deployments.

G. ETHICAL GOVERNANCE AND POLICY FRAMEWORKS

Scalable and responsible EdTech implementation of agentic AI calls for the establishment of comprehensive institutional, legal and ethical infrastructures that address the multi-faceted complexities of such technologies. Fundamental to this is data collection, usage, consent and users' rights policy making that is transparent and ensures students' personal information remains secure and ethically [105]. These policies must very explicitly detail how information are stored, passed and utilized, with users retaining control and informed authority over their presence on the web.

Also essential is an application of participatory design processes that involve major stakeholders such as students, teachers, parents and administrators directly in the development and testing of agentic AI systems. An open process raises diverse perspectives, aligns with cultural sensitivities and guarantees that technological possibilities are harmonized with actual educational needs and values. With developers' and end-users' collaboration, participatory governance can de-risk issues such as algorithmic bias, educational goal mismatch and unforeseen consequences of autonomous decision-making [106].

Globally, there is an urgent imperative for international cooperation to develop guidelines and regulatory frameworks that promote inclusive, equitable and secure adoption of agentic AI for learning worldwide. These frameworks must prioritize access and fairness such that the benefits of AI-based learning are not employed to increase the gaps or exclude marginalized groups. Transparency, accountability and auditability frameworks for AI systems will be essential in developing and maintaining public trust [107].

Lastly, ethical governance is the foundation for realizing the potential of agentic AI for educational transformation. Without robust policies and participatory design practices, autonomous learning agent use can undermine trust, perpetuate inequities and compromise learner agency.

IX. CONCLUSION

Agentic AI represents a thrilling new horizon for learning technology with the potential to enhance personalization, motivation and teaching assistance at scale. Blending breakthroughs in language models, planning, memory and multimodal interaction, these systems have the potential to transform learning experiences for diverse settings and student populations.

But bringing this into being requires overcoming the significant challenges of pedagogical fit, transparency, bias, privacy and institutional preparedness. People-centred values, ethical stewardship and scientific assessment should be the leading principles for responsible design and deployment to guarantee that agentic AI aids and facilitates educators and learners.

Human-in-the-loop procedures—when instructors monitor, revise or override agentic feedback—serve as essential safeguards to ensure AI interventions are aligned with pedagogical goals and uphold teaching integrity. Lastly, the reliability of agentic systems must be thoroughly evaluated in real-world classroom environments to ensure they are secure, coherent and pedagogically effective at scale.

In the future, research and development must focus on developing pedagogically-informed, interoperable and explainable agentic systems, fostering human-AI collaboration and constructing robust frameworks for effective and ethical integration in education. In this manner, agentic AI can contribute significantly towards offering inclusive, interesting and lifelong learning opportunities to the world.

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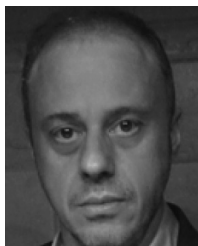
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